

RISHITHA CHILAKA

AI/ML Engineer | RAG & LLM Applications | Applied Machine Learning

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PROFESSIONAL SUMMARY

AI/ML Engineer with 4+ years of experience building and productionizing machine learning, Generative AI, and data-driven solutions. Hands-on experience developing enterprise RAG and LLM applications, retrieval pipelines, vector search, document processing, LLM evaluation, and hallucination mitigation using Python, PyTorch, Azure OpenAI, Hugging Face Transformers, and pgvector. Proven track record of improving retrieval performance, model accuracy, and production reliability through re-ranking, evaluation frameworks, prompt versioning, model routing, and monitoring. Experienced in building scalable ML pipelines and deploying AI/ML systems using Docker, MLflow, GitHub Actions, and Microsoft Azure.

TECHNICAL SKILLS

Languages & Data: Python, SQL, Bash, Git, Pandas, NumPy **Generative AI & LLMs:** Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), Re-Ranking, Vector embeddings, Semantic search, Prompt engineering, Azure OpenAI, Hugging Face Transformers, Hallucination mitigation **LLM Evaluation:** Golden datasets, Evaluation harnesses, Retrieval recall@k, Groundedness/Faithfulness evaluation, Regression testing, Error analysis, Prompt versioning **Machine Learning:** PyTorch, Scikit-learn, NLP, Document Classification, Feature Engineering, Classification, model evaluation **AI/ML systems & MLOps:** Vector Databases, pgvector, Document Processing, OCR, Retrieval Pipelines, MLflow, CI/CD, GitHub Actions, Docker, Model Monitoring, Model Versioning, Rollback **Cloud:** Microsoft Azure

PROFESSIONAL EXPERIENCE

AI/ML Engineer

Optum — Houston, TX | Jan 2025 - Present

- Designed and built production components of a Retrieval-Augmented Generation (RAG) enterprise intelligence platform using Azure OpenAI, Hugging Face Transformers, and vector embedding pipelines to support retrieval and synthesis across 40,000+ enterprise documents and structured/unstructured data.
- Built a distributed Python-based ETL and document-processing pipeline for parsing, OCR fallback, section-aware chunking, embeddings, and incremental re-indexing in pgvector, supporting ingestion and updates across a large enterprise document corpus.
- Improved retrieval Recall@10 from 0.58 to 0.79 by restructuring chunks around document section headings and adding a re-ranking stage, addressing retrieval quality as a key bottleneck to answer quality.
- Integrated a golden-set evaluation framework into GitHub Actions CI/CD to measure groundedness and Recall@K for prompt and retrieval changes, catching 3 regressions before production and establishing a repeatable quality gate.
- Reduced hallucination rate by 50% on the evaluation set by routing out-of-corpus questions to analysts instead of generating unsupported responses, improving production answer reliability.
- Reduced inference cost for simpler queries by routing single-document lookups to a smaller model and multi-document comparisons to a larger model, with evaluation showing no measurable quality degradation on the simpler workload.
- Improved document-classification F1 from 0.61 to 0.83 across 14 document classes by developing and productionizing a PyTorch based classifier, replacing keyword-rule routing for enrollment paperwork.
- Implemented prompt versioning and rollback controls after tracing a two-day answer-quality regression to an untracked prompt change, adding change control and recovery capabilities to the production AI workflow.
- Monitored production model performance and prediction quality, supporting detection of performance degradation and informing model updates.

Machine Learning Engineer

IBM — Bangalore | Jan 2020 - Jul 2022

- Developed anomaly detection models for log data processing 450K+ logs per hour, contributing to a 32% reduction in false positives.
- Developed and evaluated NLP classification models using spaCy, improving classification accuracy by 26%.
- Supported optimization of TensorFlow and CUDA-based inference pipelines to improve model response time.
- Implemented data drift monitoring processes to help maintain model stability and performance.
- Assisted with deploying and maintaining containerized ML services using Python and machine learning technologies.

EDUCATION

Union Commonwealth University — Master of Science, Management Information Systems

Aug 2022 - Dec 2024